Software Requirements Specification

For **PolyFab ERP** (Version 1.0)

Course: Software Engineering

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Table of Contents

1. Purpose..... 05

1.1. Scope..... 05

1.2. Overview..... 05

2. Overall Description..... 06

2.1. Product Perspective..... 06

2.2. Product Functionality/Features……...….06

2.3. User Characteristics..... 07

2.4. Constraints..... 09

2.5. Assumptions and Dependencies..... 09

3. Specific Requirements..... 11

3.1. Functional Requirements..... 11

3.1.1. Administrator's Requirements……11

3.1.2. Sales Person's Requirements…….……13

3.1.3. Production Manager's Requirements……15

3.2. External Interface Requirements..... 16

3.2.1. User Interfaces…….……16

3.3.2. Hardware Interfaces…….....……17

3.3.3. Software Interfaces…….....……18

3.3.4. Communications Interfaces…….....……18

4. Non-Functional Requirements..... 19

4.1. Performance Requirements.....19

4.2. Safety Requirements.....19

4.3. Security Requirements.....19

4.4. Software Quality Attributes.....19

1. Purpose

The purpose of this document is to specify the software requirements for the ERP system tailored for the polyfab industry, specifically focused on the production of polypropylene (PP) woven fabrics and sacks. This ERP system, identified as PolyFabERP Version v1.0, is designed to manage and streamline critical business operations, including sales, production, finishing, and dispatching.

1.1. Scope

Scope of the proposed system is to provide a platform where

- Order taking and tracking
- Cost Calculation
- Production monitoring
- Inventory Management
- Labour and Machinery Allocation
- Reporting

1.2. Overview

This ERP implementation encompasses the entire workflow of the polyfab industry, ensuring efficient management of sales orders, production scheduling, quality control in finishing processes, and the coordination of dispatching products to customers.

While this document provides a comprehensive overview of the system, it primarily focuses on the core modules and functionalities that are essential to the operations of the polyfab industry.

2. Overall Description

2.1. Product Perspective

The ERP system specified in this document is a new, self-contained product designed specifically for the polyfab industry, focusing on the production of polypropylene (PP) woven fabrics and sacks. As an independent solution, it will manage critical business processes, including sales, production, finishing, dispatching, and inventory management, without relying on any existing systems.

2.2. Product Functionality/Features

The ERP system for the polyfab industry includes several key modules, each designed to perform essential functions that enhance the management of polypropylene (PP) woven fabrics and sacks production. Below is a high-level summary of the key features organized by module:

2.2.1. Sales Module

- **Order Tracking:** Allows users to track the status of customer orders in real-time, ensuring timely updates and communication.
- **Price Calculation:** Automates the calculation of prices based on various parameters such as quantity, customer discounts, and promotional offers.
- **Order Taking:** Facilitates the entry and processing of customer orders through a user-friendly interface.

2.2.2. Production Module

- **Production Planning and Scheduling:** Enables effective planning and scheduling of production activities to optimize resource utilization and meet delivery timelines.
- **Waste Reporting:** Provides tools for reporting and analyzing production waste, helping identify areas for improvement and cost reduction.

2.2.3. Inventory Module

- **Raw Material Stock Management:** Manages the inventory of raw materials, ensuring adequate supply levels to support production needs while minimizing excess stock.
- **Intermediate Goods Stock Management:** Tracks and manages inventory levels of intermediate goods, facilitating better control over production flow and reducing bottlenecks.

2.2.4. Dispatch Module

- **Manage Dispatchment of Goods:** Streamlines the process of dispatching finished goods to customers, ensuring timely delivery and accurate record-keeping.

2.2.5. Admin Module

- **Production Reports:** Generates comprehensive reports on overall production activities, allowing for performance analysis and operational insights.
- **Waste Summaries for Specific Clients:** Provides tailored waste reporting for specific clients, enabling targeted analysis and feedback.
- **Quality Reports:** Tracks and reports on quality metrics to ensure products meet established standards and customer expectations.

2.3. User Characteristics

The ERP system for the polyfab industry is designed to be used by various user classes, each with specific roles and responsibilities. The following outlines the key user classes and their pertinent characteristics:

2.3.1. Admin

- **Sub-role:** None
- **Responsibilities:**
  - Create users and manage user accounts.
  - Assign waste tolerance levels to different production processes.
- **Characteristics:**
  - **Technical Expertise:** High; familiar with the overall system configuration and user management.
  - **Security/Privilege Levels:** Highest; access to all system functionalities and sensitive data.
  - **Frequency of Use:** Daily.
  - **Importance:** Highly Favoured; critical for maintaining system integrity and user management.

2.3.2. Sales Person

- **Sub-role:** None
- **Responsibilities:**
  - Take and process sales orders.
- **Characteristics:**

- **Technical Expertise:** Moderate; familiar with sales operations and order entry.
- **Security/Privilege Levels:** Medium; access to sales-related functionalities.
- **Frequency of Use:** Daily.
- **Importance:** Favoured; essential for managing customer orders and ensuring accurate order processing.

2.3.3. Manager

- **Sub-roles:** Purchase, Production, Dispatch
- **Responsibilities:**
  - **Purchase Manager:** Enter inward raw material (R/M) data.
  - **Production Manager:** Schedule production activities across different plants.
  - **Dispatch Manager:** Enter outward dispatch data for finished goods.
- **Characteristics:**
  - **Technical Expertise:** Moderate to high; varies based on sub-role.
  - **Security/Privilege Levels:** High; access to specific operational modules.
  - **Frequency of Use:** Daily.
  - **Importance:** Favoured; key to ensuring smooth operations across procurement, production, and dispatch processes.

2.3.4. Supervisor

- **Sub-roles:** Tap Plant, Loom Area, Finishing Area
- **Responsibilities:**
  - **Tap Plant Supervisor:** Start batches, manage worker shifts, enter raw material usage, record temporary production, and wastage data.
  - **Loom Area Supervisor:** Assign looms to workers, create demand for bobbins, scan bobbins for consumption, enter temporary production, and wastage data.
  - **Finishing Area Supervisor:** Track printing roll and cutting roll usage, enter temporary production, and wastage data.
- **Characteristics:**
  - **Technical Expertise:** Moderate; hands-on experience with specific plant operations.
  - **Security/Privilege Levels:** Medium; access to specific plant-related functionalities.
  - **Frequency of Use:** Daily.

- **Importance:** Favoured; vital for managing day-to-day production activities and ensuring data accuracy.

2.4. Constraints

2.4.1. Hardware Platforms:

- **Mobile Devices:** The ERP system is developed using Flutter, a cross-platform framework, making it compatible with both Android smartphones and tablets. This allows users to access the system seamlessly on mobile devices, whether on the production floor or in the field.
- **Desktop Computers:** The system includes a web application accessible via desktop computers, preferably running on the Windows operating system. This enables users to manage operations from a traditional desktop environment.

2.4.2. Operating Systems:

- **Android:** The mobile application, built with Flutter, supports the latest versions of the Android operating system, ensuring wide accessibility across a range of devices.
- **Windows:** The web application is optimized for use on Windows operating systems, providing a stable and familiar environment for desktop users.

2.4.3. Web Application:

- **Browser Compatibility:** The web application, built with a responsive design, is compatible with major web browsers like Google Chrome, Mozilla Firefox, and Microsoft Edge, ensuring smooth performance and accessibility across different user preferences.
- **Responsive Design:** The web application adjusts to various screen sizes and resolutions, offering a consistent user experience across desktops, laptops, and larger tablets.

2.5. Assumptions and Dependencies

The successful development and implementation of the ERP system rely on several key assumptions and dependencies:

Assumptions:

1. **Accurate Data Availability:**

- It is assumed that accurate and up-to-date data for all processes, including production, sales, dispatch, costing, and current production planning and scheduling, will be provided by the client. This data is critical for configuring the ERP system correctly and ensuring that it meets the operational needs of the business.

2. **Clear Procedures:**
    - Detailed and well-documented procedures for each aspect of the business (e.g., production, sales, dispatch) are assumed to be in place. These procedures are necessary for mapping business processes into the ERP system and ensuring that the software reflects the actual workflows.

Dependencies:

Third-Party Components:

- The system may rely on third-party components or services, such as Firebase for backend operations. Any changes in the availability, pricing, or functionality of these services could impact the system's operation and maintenance.

These assumptions and dependencies are crucial for the project's success. Any changes or inaccuracies in these factors could significantly affect the development, deployment, and operation of the ERP system.

3. Specific Requirements

3.1. Functional Requirements

3.1.1. Administrator

3.1.1.1. Creating a Single User

The administrator can create a single user in the system by following these steps:

3.1.1.1.1. Navigate to the "User Management" section and select the "Create User" option.

3.1.1.1.2. Provide the following details in the user creation form:

3.1.1.1.2.1. Name: Full name of the user (First Name and Last Name).

3.1.1.1.2.2. Role: Select the user's role from a predefined list (e.g., Manager, Supervisor, Salesperson).

3.1.1.1.2.3. Email: Provide a valid and unique email address for the user.

3.1.1.1.2.4. Phone: Enter a valid contact number for the user (with country code if applicable).

3.1.1.1.2.5. Department: Specify the user's department, if applicable.

3.1.1.1.2.6. Location (Optional): Provide the user's work location.

3.1.1.1.3. Assign additional details or privileges as required:

3.1.1.1.3.1. Access Level: Specify the access level for the user within their role.

3.1.1.1.3.2. Custom Permissions (if applicable): Grant any role-specific or user-specific permissions.

3.1.1.1.4. Review the details entered to ensure accuracy.

3.1.1.1.5. Submit the form to create the user.

3.1.1.1.6. Upon successful creation:

3.1.1.1.6.1. The system will display a confirmation message.

3.1.1.1.6.2. An automatic email will be sent to the user with their login credentials and instructions to access the system.

3.1.1.1.7. If there are errors during the process, the system will notify the administrator with specific error messages (e.g., duplicate email, invalid phone number).

3.1.1.1.8. Log the user creation event for audit purposes, including:

3.1.1.1.8.1. User details (excluding sensitive information such as passwords).

3.1.1.1.8.2. Administrator who created the user.

3.1.1.1.8.3. Timestamp of the creation.

3.1.1.2. Creating Multiple Users from Active Directory

3.1.1.2.1. Navigate to the "User Management" section and select the "Import from Active Directory" option.

3.1.1.2.2. Authenticate with Active Directory by providing the necessary credentials and access permissions.

3.1.1.2.3. Search for the users or user groups to be imported using filters such as:

3.1.1.2.3.1. Organizational Unit (OU)

3.1.1.2.3.2. Department

3.1.1.2.3.3. Job Title

3.1.1.2.3.4. Group Membership

3.1.1.2.4. Select the users or groups to be imported.

3.1.1.2.5. Map Active Directory fields to the system-required fields:

3.1.1.2.5.1. Name: Full name of the user.

3.1.1.2.5.2. Role: Map to corresponding roles in the system.

3.1.1.2.5.3. Email: Email address retrieved from Active Directory.

3.1.1.2.5.4. Phone: Contact number retrieved from Active Directory.

3.1.1.2.6. Validate the selected users for any missing or duplicate information:

3.1.1.2.6.1. Ensure all mandatory fields (Name, Role, Email, and Phone) are populated.

3.1.1.2.6.2. Verify that email addresses are unique within the system.

3.1.1.2.7. Confirm the import to add the users to the system.

3.1.1.2.8. Upon successful import, the system will send login credentials and instructions to each user via email.

3.1.1.2.9. In case of errors during import, the system will generate a report detailing any issues, such as:

3.1.1.2.9.1. Invalid or missing fields.

3.1.1.2.9.2. Duplicate entries.

3.1.1.2.9.3. Users already existing in the system.

3.1.1.2.10. The administrator must address the errors in Active Directory or during the mapping process and re-initiate the import.

3.1.1.3. Login

3.1.1.3.1. To login into the system, the following details are required:

3.1.1.3.1.1. Email Address – must be a valid email address

3.1.1.3.1.2. Password – Must have at least 8 characters, the text field must be in password mode.

3.1.2. Sales person

3.1.2.1. Adding Customer

3.1.2.1.1. salesperson should enter following details to add a new customer

3.1.2.1.1.1. Company name - name of the company giving order
3.1.2.1.1.2. Name - full name of customer/company representative
3.1.2.1.1.3. email - email of the customer/company representative/ company
3.1.2.1.1.4. GSTIN no. - gstin no. of the company
3.1.2.1.1.5. phone no.(optional) - phone no. of the company representative

3.1.2.2. Order Taking

3.1.2.2.1. The system shall allow salespeople to take orders from customers and companies through a sales order form with the following details.

3.1.2.2.1.1. order type - either of the self, custom, job work
3.1.2.2.1.2. salesperson - name of the salesperson taking order. It is taken by the system based on the current user.
3.1.2.2.1.3. customer details - if the order type is either custom or job work - select a existing customer or add a new customer
3.1.2.2.1.4. product - either of the bag or roll
3.1.2.2.1.5. quantity - quantity of the product in either numbers, kgs or tons
3.1.2.2.1.6. due date - order delivery date
3.1.2.2.1.7. note - note for writing additional information

3.1.2.2.2. The system shall validate all user inputs during order taking and provide error messages for invalid entries.

3.1.2.3. Price Calculation

3.1.2.3.1. The system shall display the calculated price based on configured product and quality parameters during the order-taking process.

3.1.2.3.1.1. price = unit price of the product * quantity

3.1.2.4. Inventory Integration

3.1.2.4.1. The system shall provide real-time visibility of stock availability for required products during order entry.

3.1.2.5. Order Tracking

3.1.2.5.1. The system shall allow sales personnel to track the status of orders, including:

3.1.2.5.1.1. Remaining quantity
3.1.2.5.1.2. Scheduled quantity
3.1.2.5.1.3. Dispatched quantity
3.1.2.5.1.4. Number of looms running the order

3.1.2.6. Order Management

3.1.2.6.1. The system shall allow sales personnel to generate and download a PDF of the sales order to send to the customer.

3.1.2.6.2. The system shall notify the production manager of confirmed orders and update the order status accordingly.

3.1.2.7. Order Cancellation

3.1.2.7.1. The system shall allow sales personnel to cancel an order.

3.1.2.7.2. Upon cancellation, the system shall:

3.1.2.7.2.1. Mark the order as "Cancelled."

3.1.2.7.2.2. The system shall record the cancellation reason for future reference.

3.1.2.8. Login

3.1.2.8.1. To login into the system, the following details are required:

3.1.2.8.1.1. Email Address – must be a valid email address
3.1.1.8.1.2. Password – Must have at least 8 characters, the text field must be in password mode.

3.1.3. Production Manager

3.1.3.1. Planning and Scheduling Production (Tap Plant)

3.1.3.1.1. The system shall provide a list of pending orders and orders under production, including required by date and time.
3.1.3.1.2. The system shall confirm the selected orders for production when the production manager chooses them.

3.1.3.1.3. The system shall verify the entered production quantity against the current plant capacity and raw material availability.
3.1.3.1.4. The system shall suggest adjustments (e.g., reducing quantities or removing orders) if the total quantity exceeds the plant's capacity or insufficient raw materials are available.
3.1.3.1.5. The system shall re-verify the adjusted production quantity to ensure it is within plant capacity and raw material availability.
3.1.3.1.6. The system shall create batches of products with the same parameters if all criteria are met.
3.1.3.1.7. The system shall allow the production manager to sequence batches and assign them to specific plants.
3.1.3.1.8. The system shall update the production schedule based on the assigned sequence.
3.1.3.1.9. The system shall deliver the finalized production sequence to the respective plant supervisor.

3.1.3.2. Change in Sequence Order

3.1.3.2.1. The system shall allow the production manager to reorder batches not yet in production.
3.1.3.2.2. The system shall update the sequence and notify the respective plant supervisor of any changes.

3.1.3.3. Order Cancellation for Planned Quantities

3.1.3.3.1. The system shall allow the production manager to cancel planned quantities for an order.
3.1.3.3.2. If canceled quantities are not yet under production, the system shall update the sequence and notify the respective plant supervisor.

3.1.3.4. Assignment of Looms for an Order

3.1.3.4.1. The system shall confirm the selection of an order and product by the production manager.
3.1.3.4.2. The system shall prompt the production manager to enter the number of looms to be assigned.
3.1.3.4.3. The system shall allow the production manager to select specific looms for the order.
3.1.3.4.4. Upon completing the assignment, the system shall notify the supervisor responsible for the selected looms.

3.1.3.5. Viewing Availability of Raw Materials

3.1.3.5.1. The system shall provide the production manager with a list of raw materials and their current quantities.

3.1.3.6. Login

3.1.3.6.1. To login into the system, the following details are required:

3.1.3.6.1.1. Email Address – must be a valid email address

3.1.3.6.1.2. Password – Must have at least 8 characters, the text field must be in password mode.

3.1.4. Tap Plant Supervisor

3.1.4.1. Batch Management

3.1.4.1.1. Starting a Batch

3.1.4.1.2. The system shall allow the Tap Plant Supervisor to start a new production batch.

3.1.4.1.3. The system shall record the batch start time and initialize progress monitoring.

3.1.4.1.4. Updating Raw Material Usage

3.1.4.1.5. The system shall prompt the supervisor to enter raw material usage every 30 minutes or when new raw material is added.

3.1.4.1.6. The system shall validate the entered raw material quantity against inventory availability.

3.1.4.1.7. The system shall update the inventory by subtracting the consumed raw material.

3.1.4.2. Ending a Batch

3.1.4.2.1. The system shall allow the supervisor to finalize the batch and enter wastage details.

3.1.4.2.2. If wastage exceeds predefined tolerance levels:

3.1.4.2.2.1. The system shall require the supervisor to enter a reason for excess wastage.

3.1.4.2.2.2. The system shall record the reason and finalize the batch.

3.1.4.3. Login

3.1.1.3.1. To login into the system, the following details are required:

3.1.1.3.1.1. Email Address – must be a valid email address

3.1.1.3.1.2. Password – Must have at least 8 characters, the text field must be in password mode.

3.1.5. Bobbin Marker

3.1.5.1. Bobbin Registration

3.1.5.1.1. With Barcode System

3.1.5.1.1.1. The system shall register a bobbin in inventory when the bobbin marker:

3.1.5.1.1.1.1. Attaches a barcode to the bobbin.

3.1.5.1.1.1.2. Weighs the bobbin and scans the barcode.

3.1.5.1.1.1.3. The system shall record the bobbin's weight and update inventory.

3.1.5.1.2. Without Barcode System

3.1.5.1.2.1. The system shall allow manual entry of bobbin details (weight) into inventory.

3.1.5.2. Bobbin Sorting

3.1.5.2.1. The system shall allow categorization of returned bobbins as reusable or waste.

3.1.5.2.2. Waste bobbins shall be linked to their original production batch.

3.1.5.3. Login

3.1.1.3.1. To login into the system, the following details are required:

3.1.1.3.1.1. Email Address – must be a valid email address

3.1.1.3.1.2. Password – Must have at least 8 characters, the text field must be in password mode.

3.1.6. Loom Supervisor

3.1.6.1. Roll Production

3.1.6.1.1. Starting Production

3.1.6.1.1.1. The system shall allow the supervisor to:

3.1.6.1.1.1.1. Assign 3 looms to an order.

3.1.6.1.1.1.2. Assign laborers to each loom.

3.1.6.1.1.1.3. The system shall initiate production and track loom status.

3.1.6.1.1.1.4. Production and Wastage Tracking

3.1.6.1.1.1.5. The system shall require the supervisor to enter production and wastage details every 2 hours.

3.1.6.1.1.1.6. The system shall update production logs and calculate cumulative wastage.

3.1.6.1.2. Ending Production

3.1.6.1.2.1. The system shall allow the supervisor to:

3.1.6.1.2.1.1. The system shall require the supervisor to enter final parameters (e.g., roll length, defects).

3.1.6.1.2.1.2. The system shall add the completed roll to inventory.

3.1.6.2. Bobbin Management

3.1.6.2.1. Demanding Bobbins

3.1.6.2.1.1. The system shall display required bobbin types for a loom when requested by the supervisor.

3.1.6.2.1.2. The system shall issue bobbins if sufficient stock exists; otherwise: Issue available stock.

3.1.6.2.1.3. Notify the Production Manager to prioritize production of low-stock bobbins.

3.1.6.2.2. Marking Bobbins as Used

3.1.6.2.2.1. The system shall reduce loom inventory when the supervisor scans a bobbin's barcode.

3.1.6.2.3. Returning Unused Bobbins

3.1.6.2.3.1. When an order is completed, the system shall automatically return unused bobbins to main inventory.

3.1.6.3. Pausing Production

3.1.6.3.1. The system shall mark a loom as "Paused" when requested by the supervisor.

3.1.6.3.2. The system shall record the pause start time and require a reason for the pause.

3.1.6.4. Login

3.1.1.3.1. To login into the system, the following details are required:

3.1.1.3.1.1. Email Address – must be a valid email address

3.1.1.3.1.2. Password – Must have at least 8 characters, the text field must be in password mode.

3.1.7. Dispatch Manager

3.1.7.1. Order Monitoring

3.1.7.1.1. The system shall display the production status (under production/pending/completed) for any selected order.

3.1.7.2. Dispatching Goods

3.1.7.2.1. The system shall allow the Dispatch Manager to mark goods as dispatched.
3.1.7.2.2. The system shall:
3.1.7.2.2.1. Deduct dispatched quantities from inventory.
3.1.7.2.2.2. Notify the Production Manager and Inventory Manager.

3.1.7.3. Login

3.1.1.3.1. To login into the system, the following details are required:
3.1.1.3.1.1. Email Address – must be a valid email address
3.1.1.3.1.2. Password – Must have at least 8 characters, the text field must be in password mode.

3.1.8. Purchase Manager

3.1.8.1. Raw Material Management

3.1.8.1.1. The system shall display real-time raw material stock levels with low-stock alerts.
3.1.8.1.2. The system shall update inventory when new raw materials are registered (quantity, batch, supplier).

3.1.8.2. Login

3.1.1.3.1. To login into the system, the following details are required:
3.1.1.3.1.1. Email Address – must be a valid email address
3.1.1.3.1.2. Password – Must have at least 8 characters, the text field must be in password mode.

3.1.9. Inventory Manager

3.1.9.1. Process Tracking

3.1.9.1.1. The system shall track goods dispatched for outside processing until they are returned.

3.1.9.2. Login

3.1.1.3.1. To login into the system, the following details are required:

3.1.1.3.1.1. Email Address – must be a valid email address

3.1.1.3.1.2. Password – Must have at least 8 characters, the text field must be in password mode.

3.2. External Interface Requirements

3.2.1. User Interfaces

The user interface will be designed to ensure ease of use and consistency across the application. Key characteristics include:

3.2.1.1. Screen Layout: Each screen will follow a standard layout to enhance usability. Essential elements such as navigation menus, buttons, and information display areas will be consistently positioned.

3.2.1.2. GUI Standards: The application will adhere to established GUI standards for Flutter to ensure familiarity for users.

3.2.1.3. Standard Buttons and Functions: Common buttons (e.g., Save, Cancel, Help) will be present on all screens, with keyboard shortcuts available for power users.

3.2.1.4. Error Message Display: All error messages will be clear and concise, providing users with actionable steps to resolve issues.

3.2.1.5. Screen Components: Specific screens will be developed for each user role (admin, sales person, manager, etc.) to tailor functionalities to user needs.

Note: Detailed user interface designs will be documented in a separate user interface specification.

3.2.2. Hardware Interfaces

The software product will interface with the following hardware components:

3.2.2.1. Supported Device Types: The application will support various devices, including desktop computers, tablets, and mobile devices.

3.2.2.2. Data Interaction: The system will interact with barcode scanners and printers for inventory management and dispatch operations.

3.2.2.3. Communication Protocols: The application will use standard communication protocols (e.g., USB, Bluetooth) for connecting to peripheral devices.

3.2.3. Software Interfaces

The software will interact with the following components:

3.2.3.1. Database: The application will utilize Firebase as the backend database to store and retrieve data. Version information will be provided.

3.2.3.2. Operating System: The application will be compatible with major operating systems (iOS, Android, Windows).

3.2.3.3. Data Sharing: Data such as user credentials, inventory levels, and transaction records will be shared across components. The sharing mechanism will utilize secure API calls.

3.2.3.4. API Documentation: Detailed API protocols will be referenced in separate documentation.

3.2.4. Communications Interfaces

The application will incorporate the following communication functionalities:

3.2.4.1. Email Notifications: The system will send email notifications for order updates and inventory alerts.

3.2.4.2. Web Protocols: HTTP will be used for web-based interactions, with considerations for secure communication (HTTPS).

3.2.4.3. Data Transfer Rates: Minimum acceptable data transfer rates will be defined to ensure timely processing of transactions.

3.2.4.4. Security Measures: All communications will be encrypted to protect sensitive data.

4. Non-Functional Requirements

4.1. Performance Requirements

The application must meet the following performance criteria:

4.1.1. Response Time: User actions should receive a response within 2 seconds under normal operating conditions.

4.1.2. Concurrent Users: The system should support at least 100 concurrent users without degradation of performance.

4.1.3. Data Processing: Inventory updates must occur in real-time to ensure accuracy.

4.2. Safety Requirements

Safety measures include:

4.2.1. User Training: Proper training must be provided to all users to prevent operational errors that could lead to safety hazards.

4.2.2. Regulatory Compliance: The product must comply with relevant safety regulations applicable to the manufacturing and inventory sectors.

4.3. Security Requirements

Security requirements include:

4.3.1. User Authentication: The application will implement user identity verification, requiring secure login credentials.

4.3.2. Data Protection: Sensitive data will be protected using encryption standards and hashing(e.g., AES-256, MD5).

4.3.3. Compliance: The system will adhere to relevant data protection regulations, such as GDPR.

4.4. Software Quality Attributes

Quality attributes for the software include:

4.4.1. Usability: The application should be intuitive, allowing users to perform tasks with minimal training.

4.4.2. Reliability: The system should have 99.9% uptime, ensuring that users can access the application when needed.

4.4.3. Maintainability: The software should be designed for ease of updates and enhancements.